

KnowCTI: Knowledge-based cyber threat intelligence entity and relation extraction

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ABSTRACT

Structured cyber threat intelligence enables security researchers to know the occurrence of cyber threats in time, thereby improving the efficiency of security defense and analysis. Previous works usually use general deep learning and NLP techniques to extract intelligence. Such methods suffer from insufficient semantic understanding in the field of security. To address these issues, we propose a novel method called Knowledge-based Cyber Threat Intelligence Entity and Relation Extraction (KnowCTI), which incorporates cybersecurity knowledge into the model to enhance the understanding of the realm of cybersecurity and has a full picture of threats with the threat intelligence graph generation. Specifically, we first build a cybersecurity knowledge base and train cybersecurity-aware knowledge embeddings based on the base. Secondly, we refine the most related knowledge triples by attention mechanism and gate mechanism, and then construct a sentence tree through these triples. Next, we employ graph attention networks to incorporate knowledge information into the sentence by considering the sentence tree as a graph. Finally, we consider entity extraction as a sequence labeling problem and relation extraction as a classification problem to decode the entities and relation triples according to the threat intelligence ontology we designed. Experimental results demonstrate the superior performance with the F1 score exceeding 90.16 and 81.83 on entity and relation extraction separately.

1. Introduction

The number of disclosed security vulnerabilities has been steadily increasing recently. Within the multitude of security vulnerabilities, the proportion of high-risk vulnerabilities has been consistently rising (h3c, 2023). This trend exacerbates the severity of various novel forms of vulnerability exploitation and attack methods, such as Advanced Persistent Threat (APT) attacks, posing a more pronounced impact on the cybersecurity environment (secr, 2023). Consequently, it imposes higher demands on threat detection and defense. Traditional protection methods primarily employ defensive mechanisms like firewalls and intrusion detection systems, strategically positioned at boundaries or critical nodes. However, these methods are ineffective against complex attacks and new vulnerabilities since they deeply rely on prior features, rules, and knowledge (Mitchell and Chen, 2014; Thimma et al., 2015). Researchers must consistently identify emerging vulnerabilities and threats to update their configurations. To address the limitations of current defense measures, security experts have proposed the adop-

tion of Cyber Threat Intelligence (CTI), encompassing comprehensive threat information and Indicator of Compromise (IoC) (Samtani et al., 2020). By leveraging CTI, researchers can attain a thorough comprehension of threats and transition from conventional passive defense to active defense strategies (Zhao et al., 2020).

A substantial volume of open-source threat intelligence is available on the Internet, manifesting in diverse forms across public media such as technology blogs, community forums, and social media. These sources disseminate threat-related information with extensive coverage and notable timeliness, significantly contributing to threat warning and defense efforts. Consequently, security organizations are progressively extracting CTIs from security-related text available on the Internet. This allows them to comprehensively understand existing or imminent cyber threats and develop precise defense capabilities to augment the capabilities of existing security protection equipment (Li et al., 2022), which has sparked the interest of security experts in delving deeper into research on CTI extraction.

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In recent years, numerous studies have focused on CTI extraction. The trajectory of research on CTI extraction has evolved from extracting simple intelligence to more intricate forms. Existing research can be broadly categorized into rule-based and learning-based approaches. Initial efforts primarily concentrated on rule-based methods. These methods extract IoC with regular formalities through the application of predefined rules (Burger et al., 2014; Liao et al., 2016). Nevertheless, these methods are limited to extracting basic threat intelligence, such as IP addresses and hashes, and are incapable of identifying unregulated complex entities, such as attacker names and attack methods. Additionally, certain studies employ entity dictionaries in conjunction with grammatical rules to identify complex and previously unseen security entities. These approaches heavily rely on dictionaries and lack generalizability. With the advancements in deep learning and Natural Language Processing (NLP) techniques, researchers endeavored to treat CTI extraction as an information extraction task (Long et al., 2019; Liu et al., 2022; Wang et al., 2020), aiming to extract more intricate intelligence entities such as attackers, tools, compromised organizations, etc. In addition to extracting intelligence entities, few researchers are focusing on attack graphs to achieve a global understanding of attacks (Guo et al., 2021; Li et al., 2022). The utilization of deep learning models has progressed from the initial Word2Vec+BiLSTM-based model to the current pre-trained large language model, exemplified by BERT. However, existing approaches encounter two limitations. First, whether it is Word2Vec or a pre-trained language model, they lack an understanding of the cybersecurity domain semantics. Second, current research mainly focuses on extracting intelligence entities but seldom extracts relations between entities which could provide a comprehensive overview of security intelligence.

In response to these limitations, the paper introduces KnowCTI, a knowledge-based Cyber Threat Intelligence Entity and Relation Extraction method that incorporates relevant cybersecurity knowledge into the text to enhance the comprehension of security-related content and extracts intelligence entities and relations that could be used to construct threat intelligence graphs and have a full picture of threats. In order to fuse cybersecurity knowledge, we collect structured cybersecurity data published on the Internet and extract knowledge triples to build the cybersecurity knowledge base first. Then we train cybersecurity-aware knowledge embeddings on the cybersecurity knowledge base by introducing cybersecurity semantic information. For entity and relation extraction, we design a CTI ontology that defines the concepts used in threat knowledge graphs and their relationships, providing a framework for modeling entities and relationships. Secondly, we select related knowledge triples from the cybersecurity knowledge base according to the given sentence and construct the sentence tree. Next, we consider the sentence tree as a graph and fuse the knowledge information based on graph attention networks (GAT). Leveraging the integrated knowledge information, KnowCTI extracts intelligence entities and the relation triples, which could be used to construct a threat intelligence graph to provide a comprehensive representation of security intelligence. The contributions can be summarized as follows:

- We design a knowledge ontology about cyber threat intelligence that could help operators have a full picture of cyber threats. Based on the ontology, we could construct the threat intelligence graph by extracting these entities and relations.
- We propose to incorporate external cybersecurity knowledge into the model to augment the understanding of cybersecurity text. We propose the cybersecurity-aware knowledge embedding method to introduce cybersecurity semantics to knowledge embedding. To tackle the knowledge noise issue, we propose the knowledge denoising mechanism.
- Experimental results demonstrate that our method achieves great performance. KnowCTI could achieve a 90.16 F1 score on entity extraction and an 81.83 F1 score on relation extraction. Given a report, we could generate the threat intelligence graph to give a

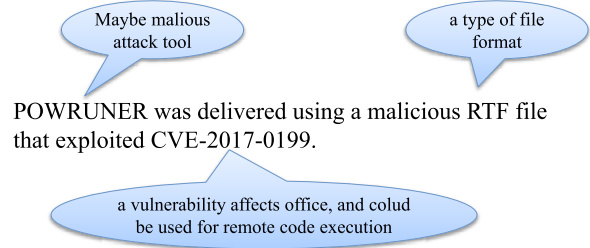


Fig. 1. An example of expert reading process.

full picture of the threat. Also, we collect a large amount of reports and explored the attack trends and attack organization profiling.

2. Background

2.1. Motivation

In the process of human reading, ordinary individuals can only comprehend words based on their context, whereas experts can make inferences with pertinent cybersecurity knowledge while reading text from the cybersecurity domain. Inspired by this, we propose the insertion of knowledge triples into security text, incorporating knowledge information to enhance the representations whose goal is to leverage domain-specific knowledge and integrate it into the text, thereby improving the understanding of cybersecurity text and ultimately empowering CTI extraction capabilities.

Experts understand which products the CVE vulnerability may affect and the attack methods attackers may employ, but models lack such an ability. For instance, in the text “POWRUNER was delivered using a malicious RTF file that exploited CVE-2017-0199”, it is describing how an attack tool named POWRUNER, utilizing the vulnerability, was transmitted to a target system through a specific file. As illustrated in Fig. 1, experts reading this text will incorporate relevant knowledge to better comprehend the text and infer that POWRUNER is an attack tool. While a cybersecurity expert finds it easy to understand such text, machines struggle as they lack knowledge about CVE and RTF. Hence, the incorporation of security knowledge into the model will significantly enhance performance. Achieving KnowCTI entails addressing the following challenges.

2.1.1. Cybersecurity-aware knowledge embedding representations

Current methods train knowledge embeddings on knowledge bases through translation-based or neural network methods (Ji et al., 2021). However, these methods are task-independent and domain-irrelevant. In other words, the embeddings generated by these methods lack cybersecurity-specific semantic information. To address this, we introduce cybersecurity-aware knowledge embedding representations for the security domain.

To incorporate security domain semantic information into knowledge embeddings, it is essential to capture the text context in which the knowledge entity is situated. For each knowledge entity, we initially conduct a search for cybersecurity reports related to it using a search engine. Subsequently, we filter sentences containing the knowledge entity. To grasp the security context, we employ an additional cybersecurity text binary classification task, training a classification model to discern whether the text is related to cybersecurity. So that we could obtain initial knowledge entity embeddings by the classification model, which would be used for training knowledge embeddings based on the translation-based method. Consequently, we could train cybersecurity-aware knowledge embeddings using cybersecurity-related reports. The detailed introduction will be provided in Section 4.2.

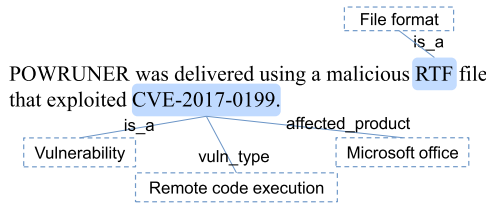


Fig. 2. Knowledge insertion and sentence tree construction.

2.1.2. Knowledge fusion into cybersecurity text

Effective utilization of knowledge positively impacts the model's performance. Conversely, improper use of knowledge can lead to counterproductive outcomes (Liu et al., 2020). The introduction of knowledge has the potential to bring knowledge noise comprising irrelevant triples or erroneous information, which may deviate from the original intended meaning of the text. An entity in the security knowledge base may possess multiple knowledge triples; however, not all of them contribute to text understanding. It is crucial for the model to select the most relevant knowledge triples to enhance text understanding.

The attention mechanism (Vaswani et al., 2017), inspired by human cognitive processes, enables models to dynamically concentrate on specific elements within input sequences, providing them with the capability to comprehend complex dependencies and generate contextually relevant and coherent language. The attention mechanism has played a pivotal role in catalyzing a paradigm shift in NLP and deep learning, particularly in the context of the Transformer architecture. For this reason, we employ attention mechanisms to denoise knowledge and fuse knowledge. Since knowledge is typically represented by triples while text is sequential, knowledge embeddings and word embeddings have distinct embedding spaces leading to the heterogeneous embedding space problem, where we solve this problem by employing linear transformation to convert them into the same space.

We first select related knowledge triples to construct the sentence tree as shown in Fig. 2 where we consider it as a graph intuitively. Then we employ the GAT (Ren et al., 2023) to incorporate the knowledge node information into the cybersecurity text. The GAT introduced a learnable attention mechanism that allocates weights between each source and target node, allowing nodes to decide which neighboring node is more important when aggregating messages from local neighbors, rather than aggregating information from all neighbors with the same weight. It can essentially choose important information fusion. The detailed introduction will be provided in Section 4.3.

2.2. Preliminaries

A singular threat indicator cannot provide a comprehensive understanding of the threat. The knowledge graph could offer a comprehensive perspective on the threat landscape, enabling operators to observe the interconnections among various entities, including threat actors, indicators, vulnerabilities, and incidents. This heightened visibility assists researchers in identifying patterns and comprehending the broader context of threats. At the core of this knowledge graph lies the ontology (Ren et al., 2022), serving as the foundational element for the structured representation and analysis of threat-related data.

The ontology is meticulously crafted to reflect the intricacies of the cyber threat intelligence domain. Entities and relations are designed to offer a structured framework for capturing, interpreting, and utilizing threat-related data. The selection of these elements draws from both industry standards and the practical needs of cyber threat intelligence professionals.

In the realm of cybersecurity, there are many standard specifications such as Common Attack Pattern Enumerations and Classifications (CAPEC) (CAPEC, 2023), Common Weakness Enumeration (CWE) (CWE, 2023), Common Vulnerabilities and Exposures (CVE) (CVE, 2023), etc. The representative specifications are STIX (STIX, 2023), and

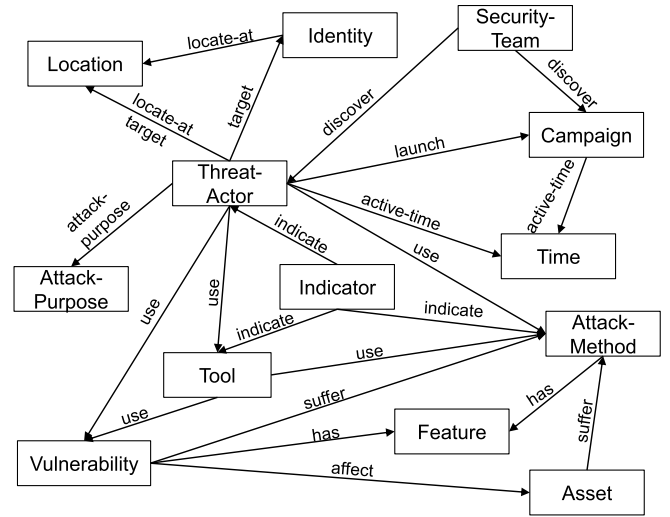


Fig. 3. Threat intelligence knowledge ontology.

UCO (Syed et al., 2016) in the field of cyber threat intelligence. These standards are crucial for cybersecurity as they enable security teams to understand and respond to threats more quickly. While constructing the ontology, we fully consider these specifications. The ontology is shown in Fig. 3.

Each threat entity contains the entity type and its definition. The knowledge ontology proposed in this study is limited within CTI. In this work, we design 13 entity types: Threat-Actor, Security-Team, Campaign, Attack-Method, Attack-Purpose, Vulnerability, Tool, Asset, Indicator, Location, Identity, Feature, Time, and the detailed explanation of each entity is shown in Table 1. Each entity type may contain several sub-classes as needed. For example, "Identity" could contain "Individuals", "Organizations", "Groups" and "Industry".

In a knowledge graph, each entity is a node and the relations between entities construct the edge. Data pertaining to knowledge graphs are expressed as (subject, relation, object) triples, denoted as (s, r, o) , signifying the existence of a relation denoted by the label r between the subject s and object o within the graph. Semantic relationships between entities play a crucial role which could provide a comprehensive understanding of a threat while isolated entities could not. Table 2 illustrates the relations we design between entities. We design 11 relation types: "locate-at", "discover", "launch", "target", "attack-purpose", "suffer", "use", "has", "affect", "indicate", "and active-time". Each relation type may contain several sub-classes as needed, for example, "attack-target" could contain "target-identity", and "target-location".

3. Overview

KnowCTI consists of 4 major modules as shown in Fig. 4: Construct Cybersecurity Knowledge Base, Cybersecurity-aware Knowledge Embedding, Knowledge Fusion, and Decoder.

To implement KnowCTI, the initial step is to construct a cybersecurity knowledge base. Fortunately, there is a wealth of structured cybersecurity data on the Internet, including sources like opencve (OpenCVE, 2023), malvuln (Malvuln, 2023), ATT&CK (ATT&CK, 2023), etc. We collect a large amount of data and extract triples to establish an initial cybersecurity knowledge base.

To ensure cybersecurity semantics for entity embedding, we introduce a cybersecurity-aware knowledge embedding training method. Initially, we obtain entity embedding representations from open cybersecurity text data containing the entity. Subsequently, we train the embedding representations on the knowledge base using translation-based methods.

In the knowledge fusion stage, we first match related knowledge triples in the knowledge base and select the knowledge triples with

Table 1
Entity type and explanation.

Entity Type	Explanation
Threat-Actor	Attacker, maybe individuals, groups, or organizations.
Security-Team	Security team, organization or institution.
Campaign	The attack incident launched by threat actor.
Attack-Method	Attack methods or ways that adversaries attempt to comprise targets.
Attack-Purpose	The purpose of the attack or campaign the threat actor launched.
Vulnerability	Vulnerability that could be used by hackers to comprise the system.
Tool	Tools used by attackers.
Asset	Assets, products or software that would be compromised.
Indicator	Indicators of comprise, such as IP, domain, hash, etc.
Location	Geographic location, which may indicate the location of the attacker or the target location of the attack.
Identity	Target individuals, organizations, groups, or industry.
Feature	The features of vulnerability, or attack method.
Time	Timestamps.

Table 2
Relationship between entities and explanation.

Subject	Relationship	Object	Explanation
Security-Team	discover	Campaign, Threat-Actor	The security team discovers an attack campaign or a threat actor carrying out an attack behavior.
Campaign	active-time	Time	Activity time of the campaign.
Threat-Actor	active-time	Time	Activity time of threat actor.
Threat-Actor	locate-at	Location	The threat actor is located at the related Location.
Threat-Actor	target	Identity, Location	The threat actor targets the type of victims and the related location.
Threat-Actor	use	Tool, Vulnerability, Attack-Method	The related attack method, vulnerability, or tool are used by the threat actor in a campaign.
Threat-Actor	launch	Campaign	Threat actor launch the attack campaign.
Threat-Actor	attack-purpose	Attack-Purpose	The attack purpose of the threat actor in the campaign.
Vulnerability	has	Feature	The characteristics of this vulnerability that would be used by threat actors.
Vulnerability	affect	Asset	Assets affected by the vulnerability.
Indicator	indicate	Threat-Actor, Tool, Attack-method	The indicator of compromise could detect evidence Threat-Actor, Attack-Pattern, malicious Tools.
Identity	locate-at	location	The identity is located at the related location.
Asset	suffer	Attack-Method	The attack method that fragile assets may suffer.
Attack-method	has	Feature	The characteristics of the attack method.
Tool	use	Attack-Method, Vulnerability	Threat actors use tools to exploit vulnerability or attack method in a campaign.
Vulnerability	suffer	Attack-Method	The vulnerability may be subject to some attack method.

similar semantics. Then we construct the sentence tree with object entities in triples as leaf nodes. Then we utilize the matrix to represent the sentence tree and use GAT to fuse knowledge information.

In the decoder stage, we employ the sequence labeling method to extract CTI entities, treating relation extraction as a classification task. In the end, we generate a threat knowledge graph according to the extracted entities and relations, facilitating a holistic view of the cybersecurity landscape.

4. Methodology

In this section, we will provide a detailed introduction to the major modules of our approach. First, we provide an introduction to the construction of the initial knowledge base from open-source structural data. Subsequently, we offer a detailed introduction to the training of cybersecurity-aware knowledge embeddings. Following that, we will describe the CTI extraction module, with a focus on knowledge fusion technology.

4.1. Construct knowledge base

Security teams frequently post summarized structural cybersecurity-related information online which represents knowledge in essence. Consequently, we can construct an initial cybersecurity knowledge base using open-source structured cybersecurity data.

The first step in constructing the cybersecurity knowledge base is to collect structured data from various sources, which provide rich and

reliable information about vulnerabilities, threat actors, attack techniques, and other cybersecurity concepts. The data sources could be divided into three types: 1) open-source database, such as OpenCVE (OpenCVE, 2023), ATT&CK (ATT&CK, 2023), CAPEC (CAPEC, 2023), etc; 2) structured data on websites, such as Malvuln (Malvuln, 2023), Threat Actor Map (aptmap, 2023, etc); 3) structured data on the open source platform github. Each data source has a specific data collection mechanism, given that we gather data from various sources.

After collecting the data, it is necessary to analyze each data format because different data sources have various data presentations. Next, we parse the collected data, extracting cybersecurity entities and their relations to generate knowledge triples. Additionally, we unify relationships from different sources that convey the same meaning. Fig. 5 shows how we extract triples from structured data of Malvuln (Malvuln, 2023) which introduces a threat AtomSilo including threat type, vulnerability type, file type, etc. We consider the AtomSilo as the subject entity, and the type content as the object entity and obtain the relations according to the type. Lastly, we identify and eliminate duplicate triples, considering that knowledge data collected from different sources may contain the same knowledge triple with varying descriptions.

4.2. Cybersecurity-aware knowledge embeddings

Training knowledge embedding representation is crucial for knowledge representation. An effective embedding representation significantly improves model performance. To incorporate cybersecurity semantics into embedding representations, we introduce an additional

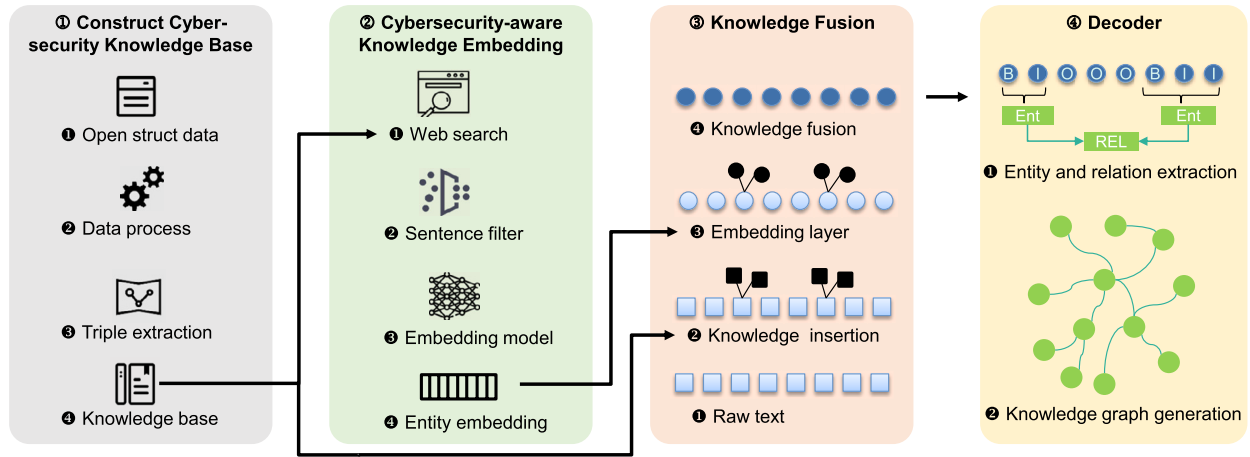


Fig. 4. The overview of KnowCTI consists of 4 modules. (1) Construct the cybersecurity knowledge base from open-source structure data. (2) Training cybersecurity-aware knowledge embeddings on the cybersecurity knowledge base by introducing cybersecurity semantics. (3) Construct the sentence tree according to knowledge triples and fuse knowledge information. (4) The decoder module which extracts entities and relation triples that could be used for knowledge graph construction.

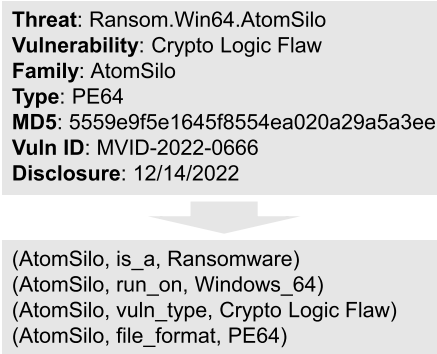


Fig. 5. Extract triples from open structured data.

cybersecurity text identification task to train the knowledge embeddings based on cybersecurity-related reports.

For a given knowledge entity e , the initial step is to search for entity-related documents P in the browser. We retain the top 10 documents assuming these are the most relevant. In most cases, sentences where the entity is located are security-related. This is because most security-related entities are unique to the realm of cybersecurity. Subsequently, we filter at most 10 sentences containing the entity as S_e to enrich the semantic information.

We train a classification model by fine-tuning BERT (Devlin et al., 2018) model consistent with the rest to determine whether the text is related to cybersecurity or not. Therefore, We could obtain entity embeddings from the finetuned model, if the text is cybersecurity-related. For each sentence, we use it as the input for the finetuned BERT model. Initially, we encode each sentence s_i and acquire the embeddings of each token. Next, we utilize the [CLS] embedding for binary classification to determine the security relevance of the sentence.

$$t_0, t_1, \dots, t_n = \text{BERT}(s_i) \quad (1)$$

$$o = \sigma(Wt_0 + b) \quad (2)$$

where t_0 is the embedding of the special token [CLS], and σ is the sigmoid activation function. If o is greater than 0.5, we think the sentence is security-related and reserves the entity embeddings. BERT uses WordPiece tokenization, which may break a word into a few tokens and generate different embeddings for each token. In this work, we only adopt the first token as the word embeddings. Also, a knowledge entity may contain multiple words. We use max-pooling to compute knowledge entity embeddings e_{emb_i} in s_i . We have multiple sentences

for each knowledge entity. Finally, we utilize the mean-pooling of e_{emb_i} to compute the final knowledge entity embeddings e_{emb} :

$$e_{emb} = f_{\text{mean}}(e_{emb0}, \dots, e_{embn}) \quad (3)$$

We make the e_{emb} as the knowledge initial representation.

Then we employ translation-based methods (Bordes et al., 2013; Wang et al., 2014; Ji et al., 2015) to train the knowledge embedding using the e_{emb} as the initial embeddings. Translation-based methods have been influential in the field of knowledge graph embedding which is simple but effective. This type of translation distance-based model usually considers the rationality of facts as the distance between two entities after translating the relationship. It uses margin loss as the object:

$$\text{loss}_{ke} = \sum_{(s,r,o) \in K} \sum_{(s',r,o') \in K'_{(s,r,o)}} [\gamma + d(s+r,o) - d(s'+r,o')]_+ \quad (4)$$

where $[x]_+$ represents the positive part of x , (s,r,o) is the triple in knowledge base K , $\gamma > 0$ is a margin hyperparameter. The corrupted triples are constructed by:

$$K'_{(s,r,o)} = \{(s',r,o) | s' \in K\} \cup \{(s,r,o') | o' \in K\} \quad (5)$$

4.3. Knowledge fusion

4.3.1. Knowledge denoise and insertion

To filter appropriate knowledge entities from a vast array of options, we employ a simple yet effective method based on security entity features. Security entities frequently possess multiple variants, each with different connectors and suffixes but sharing the same backbone string. Initially, we eliminate special characters and spaces, retain only alphabets and digits, and calculate the maximum common substring to ascertain if the text contains the target entity. We consider the ratio of the maximum common substring to the entity length, denoted as α . If α exceeds 0.8, as determined through experiments, we conclude that the text contains the target entity. Once the existence of an entity is confirmed, we utilize a string fuzzy match method to identify the closest triples and determine the insertion position. In this step, we employ an n-gram approach, where n is determined by the provided knowledge subject entity, and calculate the match score between knowledge and the target n-gram. The target entity is determined by the n-gram that achieves the highest score.

The fuzzy match method may bring incorrect knowledge triples, leading to noise. After obtaining all candidate knowledge triples, we encode the knowledge subject embeddings as $k = [k_{s0}, k_{s1}, \dots, k_{sm}]$. We

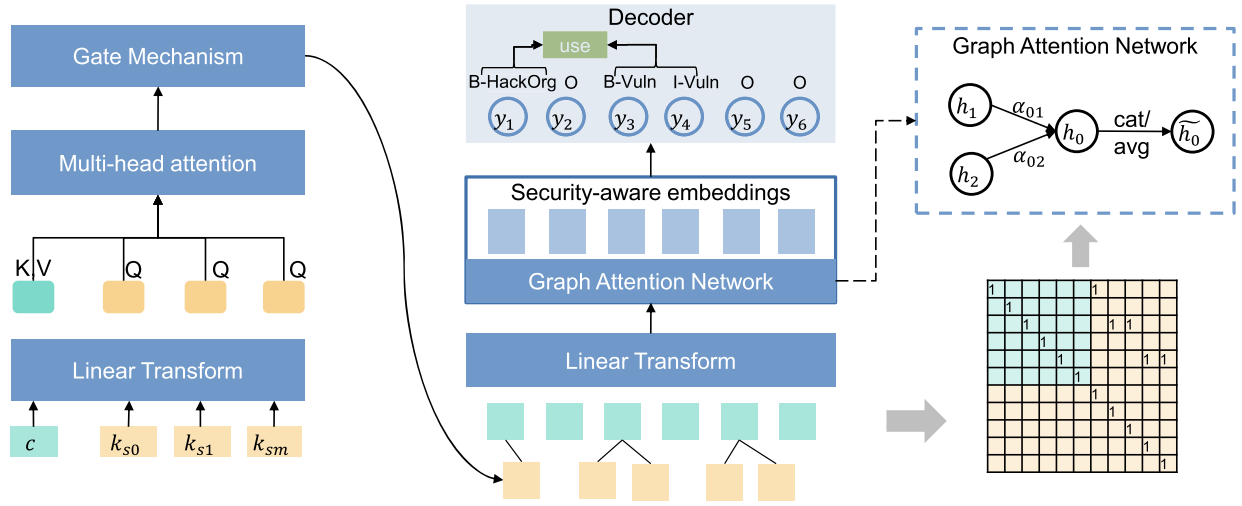


Fig. 6. Knowledge fusion architecture. We employ the graph attention networks to fuse knowledge information.

further select relevant knowledge triples by a selective module employing an attention mechanism. We evaluate the relevance of each knowledge subject entity for the target context. We encode the text with finetuning BERT model and use the special token $[CLS]$ as the context embeddings and compute the relevance. To more fully represent the features of nodes and maintain spatial feature consistency, feature transformations are performed.

$$k = kW, \quad c = cW \quad (6)$$

where W is the weight for transformation, k is the knowledge subject embedding, c is the $[CLS]$ embedding.

We employ the scaled dot-product attention function to calculate the attention score a :

$$a = \text{Attention}(QW_a, KW_a, VW_a) \quad (7)$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (8)$$

where $Q = k$ and $K, V = c$. Then we apply a gating mechanism to select the relevant knowledge and discard irrelevant knowledge:

$$\mu = \sigma(a) \quad (9)$$

In the gating mechanism, we use the sigmoid function to determine the importance of each triple. The triples with high gating values are considered relevant, while those with low values are filtered out. Then, we construct the sentence tree, according to the sentence and the knowledge triples, as shown in Fig. 2.

4.3.2. Knowledge fusion

Determining which knowledge triple to use for even the correct entities is challenging. Ideally, different knowledge triples should be applied under different semantics. Therefore, using irrelevant knowledge triples will inevitably introduce noise and reduce the precision of the threat intelligence extraction. To address this challenge, we employ Graph Attention Networks (GAT) (Velickovic et al., 2017) to integrate knowledge into text, leveraging the capability to fuse knowledge based on relevance weights between nodes.

We convert the sentence tree into a graph G , representing words and knowledge entities as nodes and the relations as edges. For a graph with n nodes, we construct the symmetric adjacency matrix $M \in \mathbb{R}^{n \times n}$ to represent the graph. If there exists an edge between node i and node j , $M_{ij} = 1$; otherwise, $M_{ij} = 0$. To preserve node information, self-loops are added to each node, represented as $M_{ii} = 1$.

As nodes in graph G are composed of words in the text and object entities in the knowledge triples, the node representation is divided into two parts. We encode the text by finetuning BERT model which is the same one used in the knowledge denoise part, and get the representation t_i of each text node; we obtain the object entity node embedding e_{emb} by cybersecurity-aware knowledge embeddings. The features of nodes are represented as $h = \{t_1, t_2, \dots, t_n, e_{emb_1}, e_{emb_2}, \dots, e_{emb_m}\}$.

To more fully represent the features of nodes and maintain spatial feature consistency, feature transformations $h_i = Wh_i$ are performed using the same weight W as in Eq. (6). Then we compute the attention weights between two nodes:

$$e_{ij} = \text{LeakyReLU}(W_g h_i, W_g h_j) \quad (10)$$

$$\alpha_{ij} = \text{softmax}(e_{ij}) \quad (11)$$

Finally, the normalized attention coefficient is linearly combined with its corresponding features to serve as the final knowledge-aware representation $\tilde{h}_i$ for each node:

$$\tilde{h}_i = \sigma\left(\sum(\alpha_{ij} W_g \tilde{h}_j)\right) \quad (12)$$

4.4. Decoder and graph construction

We consider CTI entity extraction as a sequence tagging task and consider relation extraction as a classification task and these two tasks are trained jointly. After obtaining the knowledge-aware representation of each token embedding, we pass it to a conditional random field (CRF) layer for predicting the label o_i and the highest score is considered as the prediction label:

$$\hat{y}_i = \text{argmax}(o_i) \quad (13)$$

During the training, we optimize the model by minimizing the cross-entropy loss in the named entity task and obtain the loss $loss_1$.

For the identified entities, we combine them in pairs. For each pair, we represent it by subtracting the subject entity embedding h_{ent1} from the object entity embedding h_{ent2} . As an entity may contain multiple tokens, we apply mean-pooling to the embeddings of tokens as the entity embedding h_{ent} .

$$h_r = h_{ent2} - h_{ent1} \quad (14)$$

Then we classify each pair by softmax activation after the FFNN:

$$ro_i = \text{softmax}(FFNN(h_r)) \quad (15)$$

We take the label with the highest probability as the prediction label:

$$\hat{r}_i = \operatorname{argmax}(r_{o_i}) \quad (16)$$

During the training, we also optimize the model with a cross-entropy loss in the relation classification task and obtain the loss $loss_2$.

Finally, we train the model jointly:

$$loss = loss_1 + \lambda loss_2 \quad (17)$$

where λ is the weight parameter.

5. Experiments

In this section, we introduce the datasets and conduct comprehensive experiments to demonstrate the effectiveness of our proposed model KnowCTI. In addition, we crawled more than 100,000 reports and analyzed top-k attack method trends over the past two decades. Also, we presented a case study of attack organization profiling and a case study of threat intelligence graph generation.

5.1. Datasets

We collect structural data from multiple sources and construct the initial knowledge base as introduced in Section 4.1. The total number of triples is 53,713, and the total number of entities is 14,450. We collected threat data from a set of sources automatically, including security blogs, forum posts, etc. And finally, 130,424 security-related data have been collected. To train and evaluate KnowCTI, we invested 5 cybersecurity graduate students for the annotation of the data. For security-related text classification, we annotated 4,896 instances. For CTI extraction, we annotated 8,872 instances with 28,347 entities and 13,721 triples, according to the ontology we designed.

5.2. Evaluation metrics

We use Precision(P), Recall(R), and F1 score with macro-average to evaluate the performance of entity extraction and relation extraction in KnowCTI. A triple is considered to be correctly extracted if and only if its relation type and both entities are correctly matched. Equations show the calculation method of macro-average:

$$P = \frac{TP}{TP + FP} \quad (18)$$

where TP represents true positive classifications, FP represents false positive classifications. The precision measures the capability of correct predictions among all positive predictions identified by the model.

$$R = \frac{TP}{TP + FN} \quad (19)$$

where FN represents false negative classifications. Recall indicates the capability of the model to identify positive samples.

$$F1 = \frac{2 * P * R}{P + R} \quad (20)$$

F1 score is the harmonic mean of Precision and Recall, taking into account the results of both. After obtaining the P, R, and F1 of each type, we average them to get the macro F1 score.

hit@n is commonly used to evaluate the performance of the knowledge graph embedding. It focuses on the top n predictions made by the model and assesses whether the ultimate correct entity is present within this set of top n candidates, i.e. the proportion of correct entities ranked in the top n. It could be calculated by:

$$\operatorname{hit}@n = \frac{1}{|S|} \sum_{i=1}^{|S|} \mathbb{I}(\operatorname{rank}_i \leq n) \quad (21)$$

where S is the set of triples, $\mathbb{I}$ is the indicator function (if the condition is true, the function value is 1, otherwise 0), and rank_i is the predictive ranking of the i th triple.

In this work, we use hit@10, hit@3, hit@1 as the indicator to evaluate knowledge graph embedding. The larger the hit@n indicator, the better.

MeanRank is another evaluation method of knowledge graph embedding which represents the average ranking score of correct entities. A smaller MeanRank value indicates a higher ranking and better entity vectorization results.

$$MR = \frac{1}{|S|} \sum_{i=1}^{|S|} \operatorname{rank}_i \quad (22)$$

where the symbolic meaning is the same as above.

5.3. Settings

In this work, we use the pretrained BERT model as the backbone model for fine-tuning. We use grid search to select appropriate hyperparameters. In the knowledge embedding training stage, the embedding dim is set to 768 the same as BERT, the margin is set to 10. In the CTI extraction stage, the batch size is 32. In the denoising module, we employ a multi-head attention mechanism and the number of heads is 8, and the dropout is 0.1. In the GAT module, we also use multi-head attention and set the number of heads to 8. The dropout is set to 0.5 and the α in LeakyReLU is set to 0.2. We optimize our model using the Adam optimizer with a learning rate of 5e-5. Furthermore, we incorporate a learning rate scheduler employing a cosine strategy, and we apply a weight decay of 0.01.

5.4. Main results

In order to understand the effectiveness of KnowCTI, we compare our methods with some classic methods that do not fuse knowledge. Here we give a brief introduction of them as follows:

- BiLSTM-CRF (Lample et al., 2016): It is a classic and effective method for performing NER without considering external knowledge incorporation. The BiLSTM component captures the sequential dependencies among input tokens, while the CRF component facilitates optimal and cohesive prediction of all labels within the sentence.
- BERT-CRF (Devlin et al., 2018): BERT effectively extracts implicit language information from unlabeled text, demonstrating exceptional performance across various NLP tasks, including Named Entity Recognition (NER). In this baseline, we finetune BERT on CTI extraction task, and a CRF module is applied as the top layer for inference.
- BERT-BiLSTM-CRF (Dai et al., 2019): This model combines BERT, BiLSTM, and CRF. BiLSTM layer follows by BERT, effectively capturing sequential dependencies among input tokens and enhancing the model's contextual understanding. This integrated architecture leverages the complementary strengths of each component.
- NER4CTI (Liu et al., 2022): NER4CTI is the state-of-the-art method focusing on cybersecurity threat intelligence entity extraction. It proposed a semantic augmentation approach that integrates various linguistic features to enhance the representations of input tokens and improve the NER performance.

Table 3 shows the f1 performance of different methods. The results demonstrate the effectiveness of KnowCTI as it can reach the best performance indicating that the fusion of knowledge could enhance the models' understanding of cybersecurity domain text. The results show that finetuned BERT model performs better than the BiLSTM model. The performance of the finetuned BERT model increases the BiLSTM model

Table 3

Performance of different models on CTI extraction.

Models	Entity			Relation		
	P	R	F1	P	R	F1
BiLSTM-CRF	70.04	69.98	70.01	48.35	52.68	50.42
BERT-CRF	78.87	77.66	78.26	64.95	66.06	65.50
BERT-BiLSTM-CRF	80.55	80.27	80.41	68.92	67.83	68.37
NER4CTI	84.53	84.23	84.38	-	-	-
KnowCTI	92.96	87.52	90.16	78.43	85.54	81.83

by 8.15. Because BERT is pretrained on a large corpus having better presentation ability and it is also easy to finetune to the downstream task. In KnowCTI, we use BERT as the backbone model. Compared to the BERT-BiLSTM-CRF model, KnowCTI has increased by 9.75 performing best. Because BERT is pretrained on the general corpus lacking cybersecurity semantics, the introduction of external knowledge could augment the cybersecurity semantics which could enhance the representations of cybersecurity text and then improve the model's performance. We also compare the entity extraction performance with NER4CTI, and increase by 5.78. Results show that the performance of relation extraction is worse than entity extraction generally, perhaps due to error propagation. Because entity extraction errors will inevitably lead to incorrect relation classification.

5.5. Ablation study

To understand the contribution of different modules, we conducted several ablation studies. In this section, we studied the role of knowledge fusion and the effect of different Knowledge representations.

5.5.1. Knowledge embedding evaluation

In this section, we study the performance of different translation-based knowledge embedding methods and the effectiveness of cybersecurity-aware knowledge embeddings. We compare the TransE, TransD, and TransH methods that train on knowledge base only, and the cybersecurity-aware TransE (Abbreviated as Cyber-TransE), cybersecurity-aware TransD (Abbreviated as Cyber-TransD), cybersecurity-aware TransH (Abbreviated as Cyber-TransH) that train on knowledge base using the cybersecurity entity embeddings. We evaluate the performance of knowledge embedding, and the results are shown in Table 4. Then for each method, we evaluate the performance of CTI extraction and choose the best of them as the final method we use. The results are shown in Table 5. It should be noted that the F1 score for security-related classification during training the cybersecurity-aware knowledge embeddings reached 98.68.

As shown in Table 4, we find the cybersecurity-aware translation-based methods achieve a little better performance generally. The Cyber-TransD has the best performance since it introduces more parameters to map vectors. On the other hand, this method also has the highest cost. Results in Table 5 show that the Cyber-Trans methods achieve better performance on CTI extraction than the Trans methods which demonstrates the effectiveness of the cybersecurity-aware knowledge embedding. The average improvement of Cyber-Trans methods on CTI entity extraction is 0.42, and 1.08 on CTI relation extraction. The Cyber-Trans methods only have a slight improvement over the Trans methods on CTI entity extraction and the main improvement of Cyber-Trans methods lies in precision. We have found that the improvement in the precision of CTI entity extraction will lead to an increase in the recall rate of relation extraction. Also, the method that has better performance on knowledge embedding could achieve better performance on CTI extraction. We could see that the Cyber-TransD method gets the highest score 81.75 on hit@10, 170.02 on MR and it also gets the highest f1 score on CTI extraction 90.16 on entity extraction, and 81.83 on relation extraction.

Table 4

The performance of different knowledge embedding methods.

Methods	MR	hit@10	hit@3	hit@1
TransE	215.41	81.00	74.25	60.80
Cyber-TransE	215.91	81.30	74.50	61.05
TransH	221.96	81.05	74.55	60.05
Cyber-TransH	206.03	81.25	74.65	61.50
TransD	218.60	81.15	74.50	60.90
Cyber-TransD	170.02	81.75	74.75	62.35

Table 5

The performance of different knowledge embedding methods on CTI extraction.

Methods	Entity			Relation		
	P	R	F1	P	R	F1
TransE	90.34	88.93	89.32	77.47	83.00	80.14
Cyber-TransE	91.63	87.99	89.61	76.57	86.25	81.12
TransH	90.68	88.32	89.48	77.80	82.41	80.04
Cyber-TransH	91.17	88.24	89.68	78.41	85.24	81.68
TransD	90.32	88.46	89.38	79.10	83.42	81.20
Cyber-TransD	92.96	87.52	90.16	78.43	85.54	81.83

Table 6

Performance of module deletion.

Methods	Entity			Relation		
	P	R	F1	P	R	F1
KnowCTI	92.96	87.52	90.16	78.43	85.54	81.83
wo knowledge denoise	87.80	85.02	86.39	77.73	81.48	79.56
wo fuzzy match	81.72	77.11	79.35	22.55	24.46	23.47
BERT	78.87	77.66	78.26	64.95	66.06	65.50

5.5.2. Knowledge fusion performance

According to the experiment results in Table 5, we choose Cyber-TransD method in KnowCTI. In this section, we conduct an ablation study to understand the effectiveness of each module in knowledge fusion, and the results are shown in Table 6. We remove each of them to observe their impact on the model.

First, we remove the knowledge denoise module, which refines the knowledge triples that are inserted into the text, and the performance drops by 3.61. The results show that noisy knowledge triples will drop the performance of the model. Second, we remove the fuzzy search module and use the strict match search method. This drops a lot and gets the worst even worse than the base BERT model on relation extraction as shown in the results. It may be because open texts are casual, the strict knowledge is not often matched in the open text, causing the knowledge fusion module not to learn well.

5.6. Case study

In this section, we collect 130,424 open reports from 2002-2022 and the number of reports every year are shown in Fig. 8. Then we employ KnowCTI to extract entities and relation triples from the reports. Finally, we extract 2,074,149 entities and 912,521 relation triples. From the extracted entities and relation triples, we could conduct statistical analysis, explore security knowledge, and generate threat graphs.

5.6.1. Statistical analysis of attack methods

In this part, we analyzed the extracted entities, mainly focusing on the most commonly used attack methods. Among the extracted entities, the top-5 attack methods are dos/ddos, phishing, malware, sql injection, ransomware. Then we statistically analyzed the changes in attack trends every year, as shown in the Fig. 7. In addition, due to the popularity of supply chain attacks in recent years, we have also conducted a statistical analysis of supply chain attack trends.

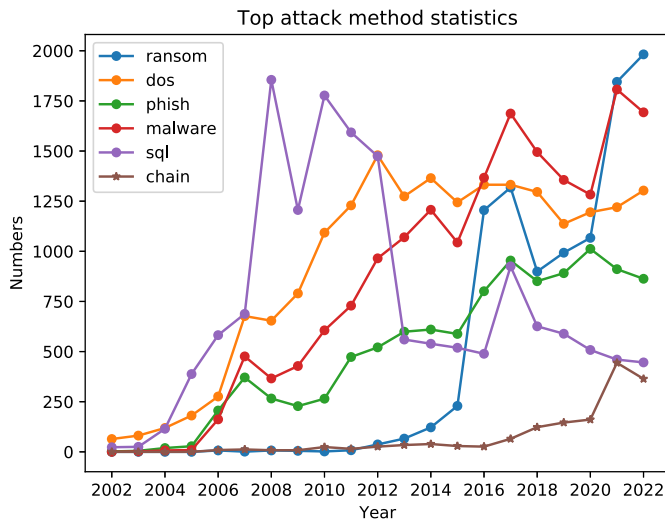


Fig. 7. Statistics of attack trends. The horizontal axis represents the year, and the vertical axis represents the reported quantity.

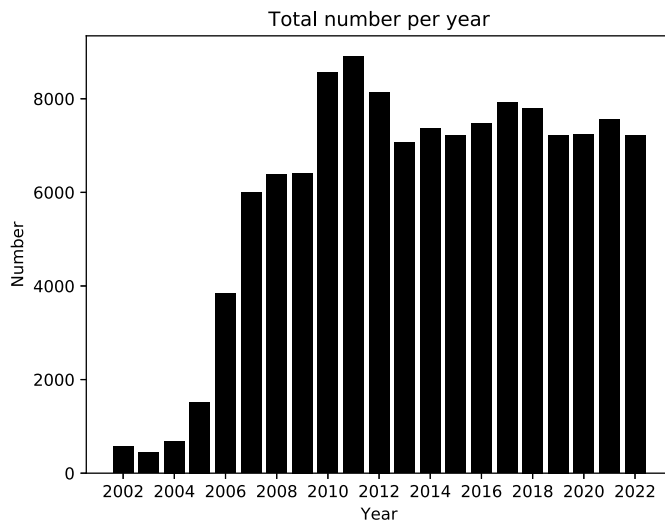


Fig. 8. Total reports per year from 2002 to 2022.

According to statistical data, we found that dos/ddos attacks began to become popular after 2010 and have continued to this day, while ransomware attacks began to gradually emerge after 2010 and rose after 2015. SQL injection attacks were quite popular from 2008 to 2012 and the number of sql injection attacks has decreased in recent years. Malware attacks and phishing attacks are on the rise as a whole, and the number of attacks will be the highest in 2020-2022, which may also be related to the COVID-19 epidemic with the increasing demand for on-line office work. The number of supply chain attacks is relatively small, but with the outbreak of the SolarWinds incident in 2021, supply chain attacks have also become hot, and the number of related attack reports has reached its peak in 2021.

5.6.2. Threat intelligence graph

We randomly take two open reports (Nelson, 2022; scmagazine, 2023) as examples to display the effectiveness of our method. For each report, we utilize KnowCTI to extract threat entities and relations and generate a threat intelligence graph. On the other hand, we construct the graph manually, which is extracted by 5 members of our team and integrated together. As shown in Fig. 9, the left part is generated by KnowCTI, and the right part is constructed by team members. In the report (Nelson, 2022), KnowCTI identifies “CVE-2021-36260”

and “vulnerabilities” as Vulnerability type, “Hikvision surveillance cameras” and “Hikvision cameras” as Asset type, “command injection flaw” as Attack-Method, “Chinese” and “Russian” as Location, “MISSION2025/APT41, APT10” and “threat actor groups” as Threat-Actor. KnowCTI identifies the relation triples as (‘CVE-2021-36260’, ‘affect’, ‘Hikvision cameras’), (‘Hikvision surveillance cameras’, ‘suffer’, ‘command injection flaw’), (‘MISSION2025/APT41, APT10’, ‘locate-at’, ‘Chinese’), (‘MISSION2025/APT41, APT10’, ‘use’, ‘vulnerabilities’), (‘threat actor groups’, ‘locate-at’, ‘Russian’), (‘threat actor groups’, ‘use’, ‘vulnerabilities’). We could extract the entities and relations effectively. The “Hikvision surveillance cameras” and “Hikvision cameras” describe the same entity; “CVE-2021-36260” and “vulnerabilities” both point to the same entity. We align the entities by rules and construct the graph according to the relation triples. Another case is shown in Fig. 10. KnowCTI extracts the Threat-Actor entity “W3LL”. From the graph, we can see that W3LL has been active since 2017 until now. It mainly uses phishing kits to launch attacks on Microsoft 365 business email accounts in region United States, Australia, and Europe, using proxy attacks and bypassing MFA attack methods. The cases demonstrate that the graph generated by KnowCTI could give a global picture of threats, and also demonstrate the effectiveness and generalization of our method.

5.6.3. Attack organization portrait

Since we have extracted a large number of relation triples, we can mine information about attack organizations and profile them. Taking the hack organization “DarkHotel” as an example, for every triple possessing an identical relation to that of the DarkHotel subject, the relevant objects are merged into a list. The extracted relation triples related to DarkHotel include attack targets, attack methods, active time, attack purpose, and tools they used. After deduplication, we summarize the organization information in Table 7.

We could see that DarkHotel has been active since 2007 and is still active today. The main attack targets of DarkHotel are senior executives and luxury hotels where we find that there are many related entities such as “business executives”, “traveling executives”, and “luxury hotels”, etc, in the extracted triples that have target relation. In addition, since the occurrence of COVID-19, the health organization has also become the target of Darkhotel. Darkhotel mainly uses spearphishing attack methods to deliver malicious code. According to the tools they used, we know that they may inject the hotel wifi networks and deliver malware, spyware, or malicious documents. And the main attack purpose is to steal information. We compared our profile results with Wikipedia data entries (Wikipedia, 2023) and found that our profile is generally correct. This also confirms the effectiveness of our method.

6. Limitation and future work

6.1. Entity alignment

KnowCTI mainly performs threat intelligence entity and relation extraction tasks, without entity alignment function. In this work, we employ simple entity alignment methods when constructing the threat intelligence graph. However, the ability of this method is relatively limited. We will address this issue in the future. Entity alignment is crucial for knowledge graph construction. In the field of NLP, the goal of entity alignment is to link entities from different knowledge graphs or databases that refer to the same real-world objects or concepts, even when these entities are expressed in different languages or come from different sources. We have extracted a large number of entity and relation triples, and after entity alignment, we can construct a large-scale intelligence knowledge graph that can be used to comprehensively understand the threat ecosystem and make threat predictions (Rastogi et al., 2021).

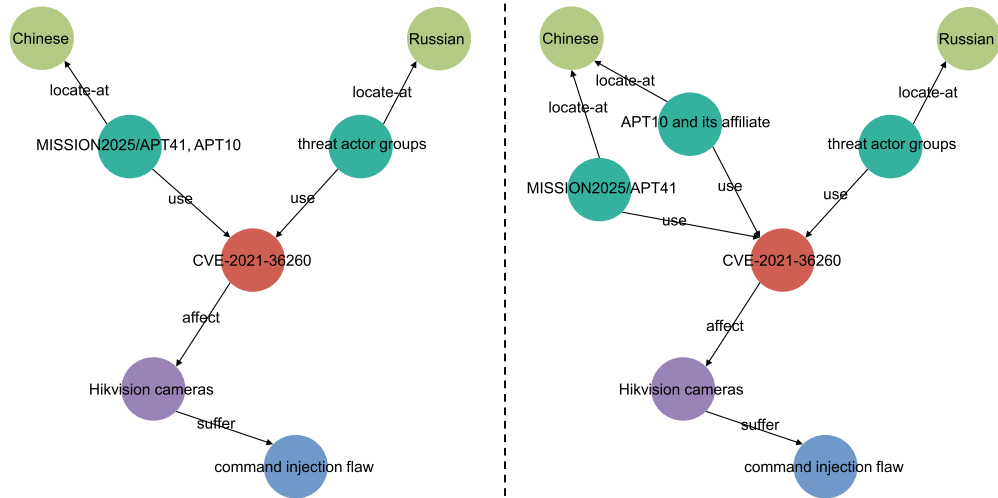


Fig. 9. A case in which KnowCTI generates the knowledge graph from the report in which disclosed attackers exploited the CVE-2021-36260 vulnerability to access Hikvision cameras.

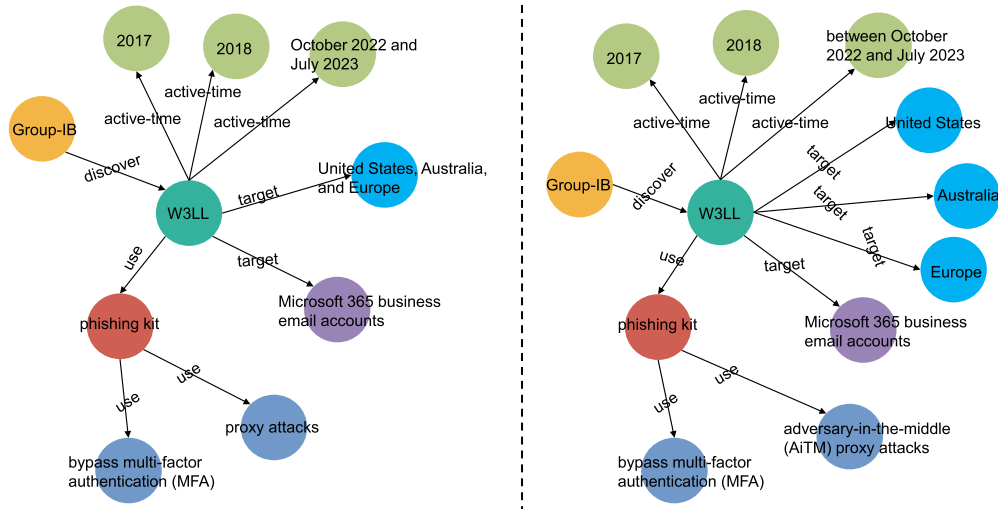


Fig. 10. KnowCTI generates the intelligence graph of the report titled “Microsoft 365 accounts targeted by W3LL threat group”.

Table 7

The portrait information of the DarkHotel attack organization.

DarkHotel	
Information	Details
active time	2007, 2013, 2014, 2016, March 2018, 2020, 2022
attack purpose	steal information, cyber espionage campaign
attack method	uploading malicious code, spearphishing
attack target	executives, luxury hotels, telecommunications, Health Organization, agencies
tools	WiFi networks, Konni malware, P2P, malware, office documents, keylogger, Adobe Flash, zero-day vulnerability

6.2. Deep analysis and targeted defense

Intelligence is valuable for targeted network defense, so further data analysis and mining are necessary. Intuitively, according to the attack method trends, security researchers could strengthen defense in a targeted manner to reduce the risk of attack. Attacks caused by vulnerability exploitation are usually related to specific products or software. If there are such products or software in the system, targeted defense is needed. Otherwise, there is no need to pay attention to this intelligence. Therefore, the extracted asset entities related to vulnerabilities can be used for targeted defense. Furthermore, the extracted vulnerability features are important for automated targeted defense (Feng et al.,

2019). We will explore automated intelligence-driven security strategy generation and development in future work, including access control configuration, attack detection rules, etc.

7. Related work

7.1. Knowledge fusion work

Incorporating knowledge information to enhance word representations (Peters et al., 2019; Liu et al., 2020; Yuan et al., 2021; Faldu et al., 2021) and augment the performance of downstream tasks (Chawla et al., 2021; He et al., 2020; Nie et al., 2021; Agarwal et al., 2023)

has been a hot research topic in NLP. KnowBERT (Peters et al., 2019) is an extension of BERT aiming to enhance word representations. It proposes to incorporate world knowledge bases into BERT by designing the knowledge attention and recontextualization component. It uses a TransformerBlock to fuse knowledge and uses another multi-head attention to recontextualize the word piece representations. K-BERT (Liu et al., 2020) is an improved BERT that enables injecting knowledge into models. It injects knowledge triples into the sentence and converts it into a sequence with a visible matrix preserving structural information and soft-position mechanism so that it could easily use BERT architecture. KeBioLM (Yuan et al., 2021) is a biomedical knowledge-aware pretrained language model, which incorporates medical knowledge from Unified Medical Language System. It first encodes text only and then encodes text-entity fusion based on Transformers. KARL-TransNER (Chawla et al., 2021) incorporates world knowledge for NER based on Transformer Encoder. It uses a self-attention network to encode entities and relations in the knowledge base. And then generate the contextualized representations for feature augmentation to improve the NER performance. KAWR (He et al., 2020) proposes to encode external knowledge into word representations for NER. It proposes a gated entity-based recurrent unit based on a recurrent neural network unit to encode entity information into the word and get entity-based word representations. KaNa (Nie et al., 2021) is a knowledge-aware NER framework that aims to denoise the type-heterogeneous knowledge in world knowledge to improve NER performance.

7.2. Cyber threat intelligence extraction

Nowadays, security researchers attempt to extract threat intelligence entities from open-source reports automatically so that they can keep up with the rapidly evolving landscape of threats and prepare for network defense in advance (Mu et al., 2018; Wang et al., 2023; Zhao et al., 2020). iACE (Liao et al., 2016) is an IOC extraction tool that utilizes regex and terms to extract IOCs and employs the sentence dependency graph to mine related tokens and relations between them from open articles. iACE relies on expert knowledge. Long et al. (2019); Zhou et al. (2018); Liu et al. (2022) apply end-to-end neural-based sequence labeling to IOC identification considering it as a named entity recognition task. The neural-based methods could identify complex entities without human hand-crafted efforts. NER4CTI (Liu et al., 2022) aggregates multiple features to augment token representations including morphological features, constituent features, and POS features. Moreover, it utilizes similar words in the cybersecurity domain corpus to augment domain semantics. Recently, work trends towards constructing graphs. AttackKG (Li et al., 2022) designs technique templates to extract attack behavior graphs for CTI reports. TTPdrill (Husari et al., 2017) designs the threat action ontology and extracts TTPs using multiple NLP techniques from CTI reports. CyberRel (Guo et al., 2021) employs the BERT-BiGRU model to extract entity and relation jointly for cybersecurity concepts which could be used to construct the cybersecurity knowledge graph. CSKG4APT (Ren et al., 2022) focuses on the APT attack scenarios and extracts CTI entities and relations to construct the knowledge graph based on the design ontology.

8. Conclusion

In this paper, we propose to integrate cybersecurity knowledge into the CTI extraction model based on pretrained language model to augment the understanding of cybersecurity semantics. Furthermore, we design a threat ontology for the threat knowledge graph and extract entities and relations, which could be used for threat knowledge graph construction and give a comprehensive picture of threats. Firstly, we build the initial knowledge base from open-source structured data. Furthermore, we train the cybersecurity-aware knowledge embeddings, which introduces the cybersecurity semantics from security texts. Secondly, we propose to fuzzy match related knowledge triples and select

more accurate knowledge triples by an attention mechanism with a gate. Based on the triples, we construct the sentence tree. Thirdly, we utilize the graph attention networks to fuse knowledge information. Finally, we extract entities by sequence labeling and extract relations by entity pair classification. Experiments on the dataset demonstrate the superior performance of KnowCTI. We have extracted CTI entities and relations from a large number of open texts and studied the activity of popular attack method trends in the past two decades. Based on the extracted relation triples, we studied the portrait of attacking organizations and threat intelligence knowledge graph generation demonstrating the effectiveness and generalization of our method.

CRedit authorship contribution statement

Gaosheng Wang: Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Peipei Liu:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Jintao Huang:** Writing – review & editing, Visualization, Investigation, Data curation. **Haoyu Bin:** Writing – review & editing, Software, Resources, Data curation. **Xi Wang:** Writing – review & editing, Project administration. **Hongsong Zhu:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hongsong Zhu reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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